

Lecture 8

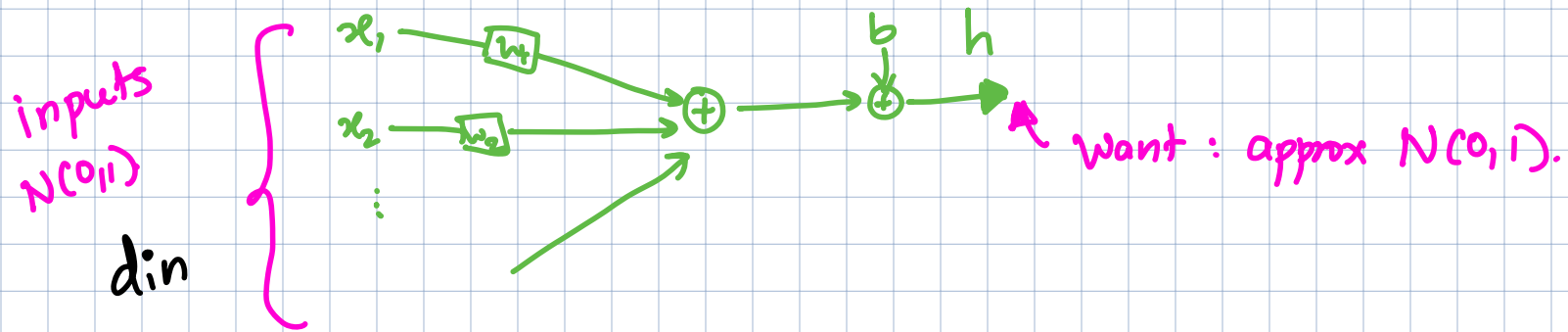
- RMS $\rightarrow$ RMS induced norm.
- Muon optimizer
- Maximal update parameterization (μP).

Reminder:

Quiz 1 tomorrow!

So what might be a good norm for deep learning?

Recap Xavier initialization.



Can we rewrite this in terms of norms?

What norm are we trying to preserve?

$$h = \vec{w}^T \vec{x} = \sum w_i x_i$$

$$E[w_i^2] = \sigma_w^2$$

$$w_i \text{ i.i.d. } E[w_i] = 0$$

$$E[h^2] = \sigma_w^2 [\sum x_i^2]$$

$$\text{if } x_i^2 \approx 1 \text{ we want } \sigma_w^2 \approx \frac{1}{d_{in}}$$

Consider RMS-norm:

$$\|\vec{x}\|_{\text{RMS}} = \frac{1}{\sqrt{d_{in}}} \|\vec{x}\|_2 = \frac{1}{\sqrt{d_{in}}} \sqrt{\sum_i x_i^2}$$

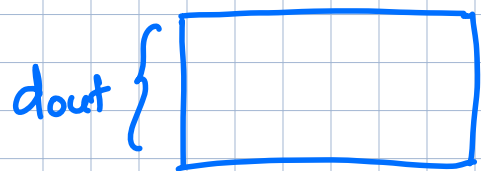
① Recall, induced matrix norm:

$$\|A\|_{\alpha \rightarrow \beta} = \max_{\|\vec{x}\|_{\alpha} = 1} \|A\vec{x}\|_{\beta}$$

RMS $\rightarrow$ RMS norm $\rightarrow$ Motivated by Xavier initialization.

$$\|A\|_{\text{RMS} \rightarrow \text{RMS}} = \max_{\|\vec{x}\|_{\text{RMS}} = 1} \|A\vec{x}\|_{\text{RMS}}$$

$$A \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$$



$$A\vec{x} = \vec{h}$$

$\uparrow$ $\mathbb{R}^{d_{\text{in}}}$ $\uparrow$ $\mathbb{R}^{d_{\text{out}}}$

$$= \max_{\|\vec{x}\|_2 = \sqrt{d_{\text{in}}}} \frac{1}{\sqrt{d_{\text{out}}}} \cdot \|A\vec{x}\|_2$$

$$= \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \max_{\|\vec{x}\|_2 = 1} \|A\vec{x}\|_2$$

$$= \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|A\|_2$$

Spectral norm
 $\|A\|_2$!

② Recall, optimizer recipe.

$$\operatorname{argmin}_{\|\Delta w\| \leq \eta}$$

Choose appropriate norm.

$$\langle \underbrace{\nabla_w L(w)}_{\text{Gradient of Loss}}, \underbrace{\Delta w}_{\text{change in } w} \rangle.$$

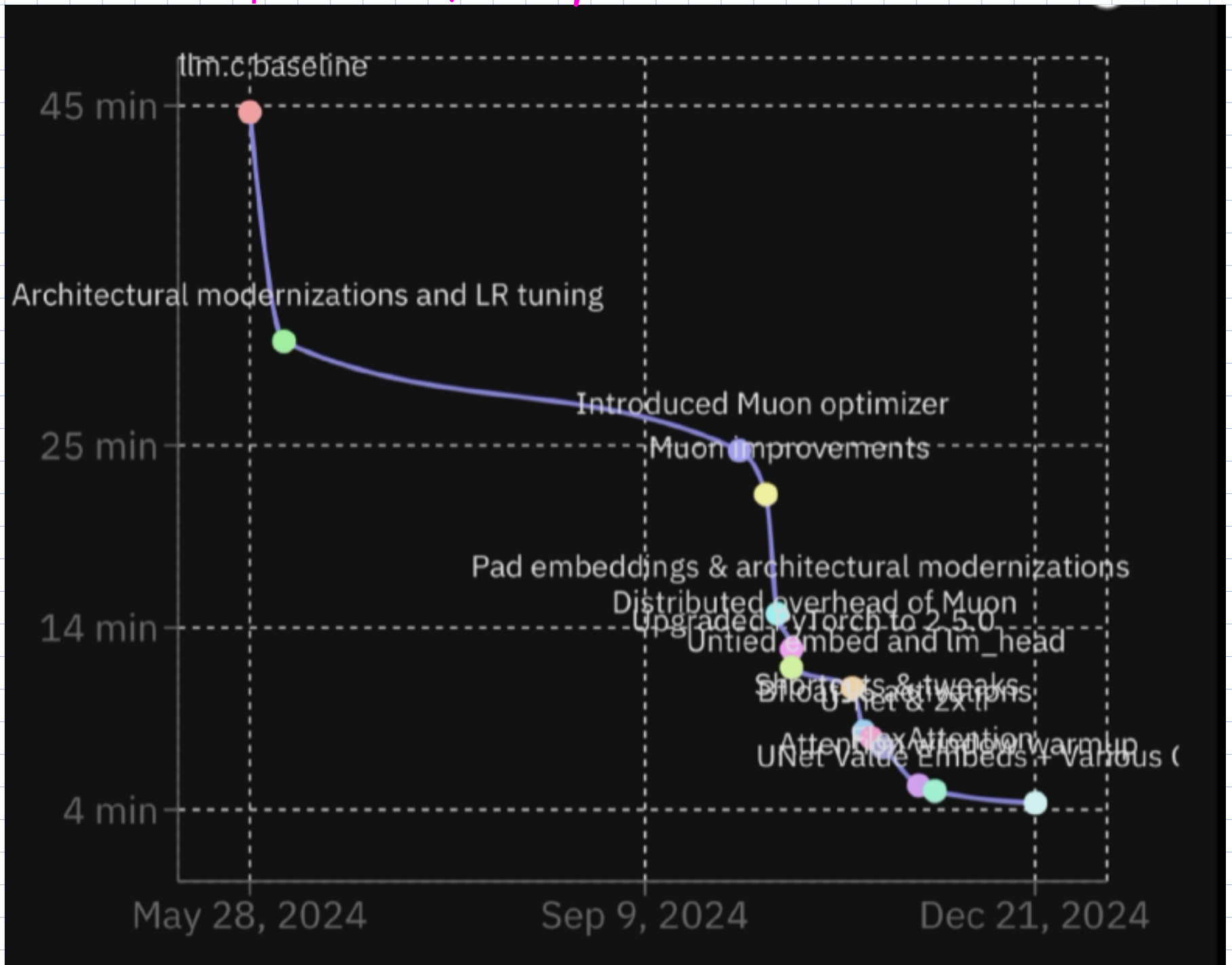
→ Choose norm

→ Choose η

↑
requires sweep!

⇒ Get optimizer.

Nano GPT speedrun.



What if we had the RMS-RMS norm instead?

$$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$$

$$\underset{\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \eta}{\text{argmin}} \quad \langle \nabla_W \mathcal{L}(W), \Delta W \rangle$$

$$= \underset{\|\Delta W\|_2 \leq \sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} \cdot \eta}{\text{argmin}} \quad \langle \nabla_W \mathcal{L}(W), \Delta W \rangle$$

Muon key idea 1.

$$\Rightarrow \Delta W^* = -\eta \cdot \sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} U_r V_r^T$$

Why might semi-orthogonal matrices be a good idea?

→ Condition number $U_r V_r^T = 1$. Uniform step in all directions.

→ Not dominated by largest singular values.

But computing UV^T is expensive.

→ Needs SVD.

Muon solves this issue through two observations

(1) Getting the direction approximately correct is good enough.

(2) Newton-Schulz iterations. ← Muon key idea 2.

$$A = \underline{U\Sigma V^T} \longrightarrow \underline{UV^T}.$$

i.e. replace all the singular values by 1.

How?

$$f(U\Sigma V^T) \approx UV^T$$

Newton-Schulz.

$$p(x) = a_1 \cdot x + a_3 \cdot x^3 + a_5 \cdot x^5 + \dots$$

① Odd polynomials commute with the SVD.

$$p(x) = a_0 x + a_1 x x^T x + a_2 (x x^T)^2 x + \dots + a_n (x x^T)^n \cdot x.$$

e.g. $p(x) = \frac{3}{2} \cdot x - \frac{1}{2} x x^T x$.

$$\begin{aligned} p(U \Sigma V^T) &= \frac{3}{2} U \Sigma V^T - \frac{1}{2} (U \Sigma V^T)(V \Sigma^T U^T) \cdot U \Sigma V^T \\ &= \frac{3}{2} U \Sigma V^T - \frac{1}{2} U \Sigma \Sigma^T \Sigma V^T \\ &= U \left[\frac{3}{2} \Sigma - \frac{1}{2} \Sigma \Sigma^T \Sigma \right] V^T \\ &= U p(\Sigma) \cdot V^T \end{aligned}$$

So $p(x) = U \underline{p(\Sigma)} V^T$.

So can apply to a matrix without changing the singular vectors,

② Can we find $p(x)$ such that
 $p(x) \rightarrow 1$ for $x > 0$

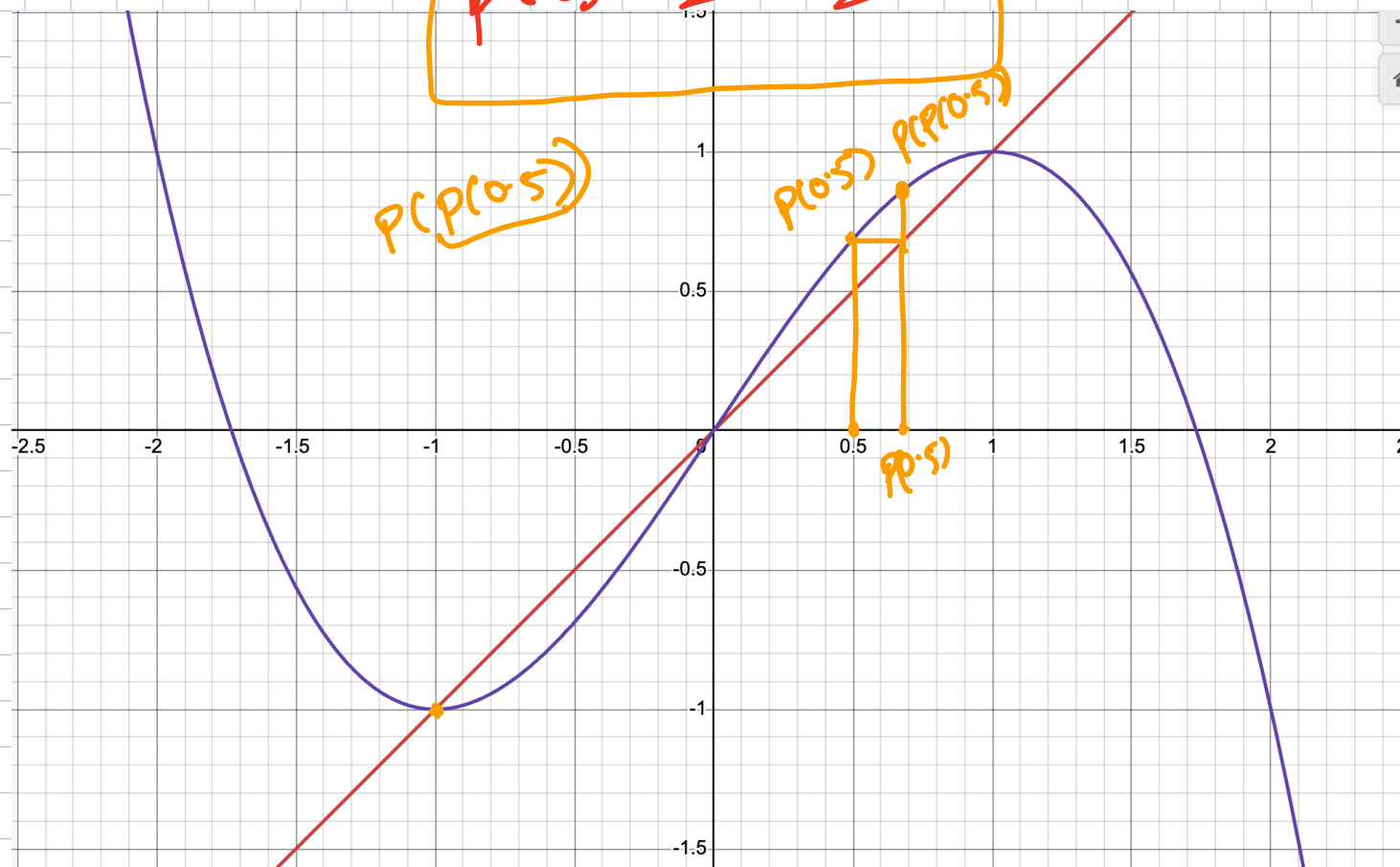
$$p(x) \rightarrow U p(\epsilon) V^T$$

$$p(p(x)) \rightarrow U p(p(\epsilon)) V^T$$

$$\vdots$$

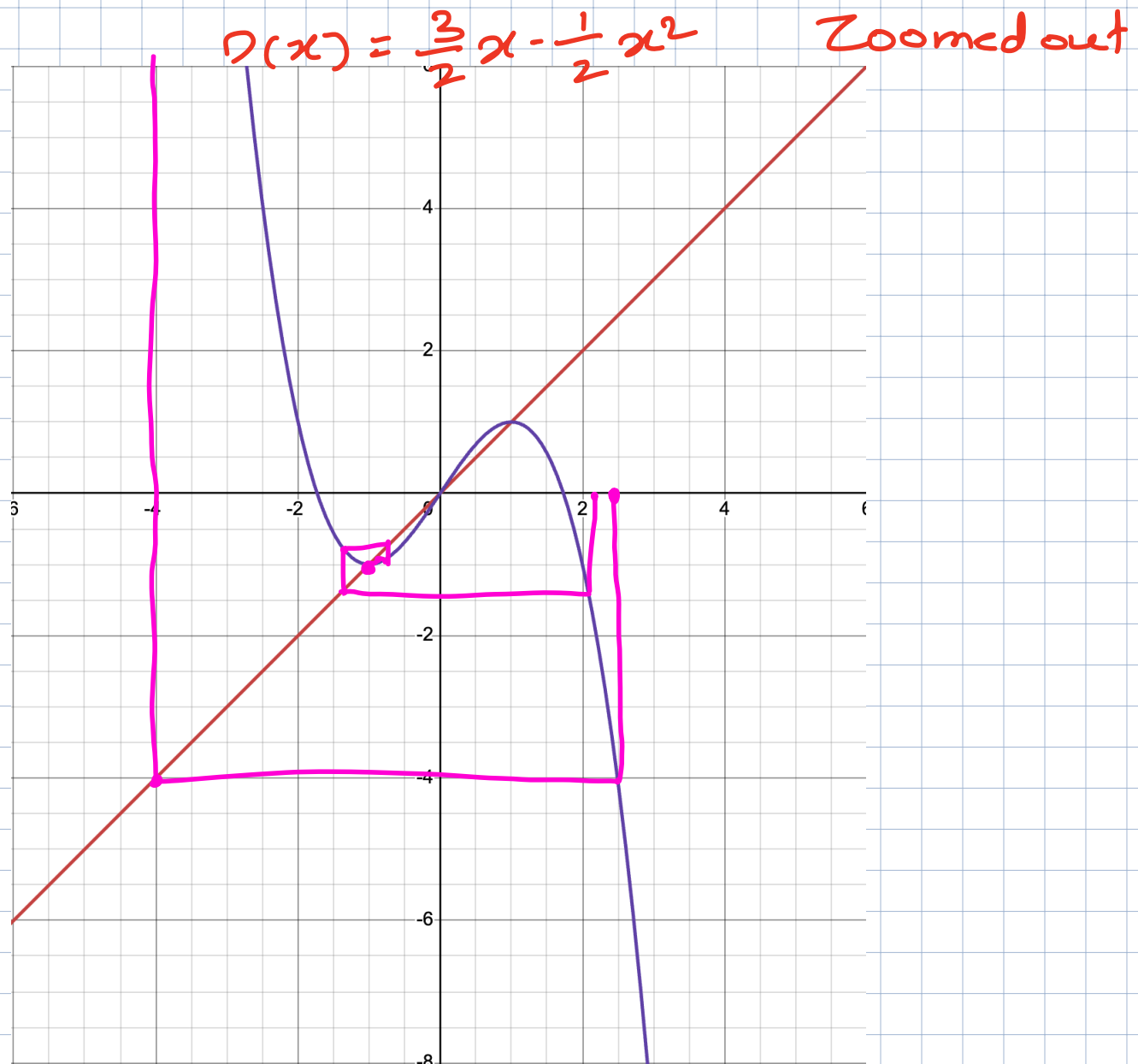
Then, iteratively apply p .

$$p(x) = \frac{3}{2}x - \frac{1}{2}x^3$$



$$p(p(p(p(\sigma)))) \rightarrow p^n(\sigma) \rightarrow 1$$

$$n \rightarrow \infty$$



→ Need to have singular values $\in (0, 1]$ before applying!

→ How? Normalize by the Frobenius norm : $\frac{G}{\|G\|_F}$.

Muon (Momentum Orthogonalized by Newton Schulz)

$$B_t = \mu \cdot B_{t-1} + \nabla_w \mathcal{L}(w)$$

$$O_t = \text{NewtonSchulz}(B_t)$$

$$W_t = W_{t-1} - \eta \cdot O_t$$

How to choose $p(x)$?

$$p(x) = ax + bx^3 + cx^5 + \dots$$

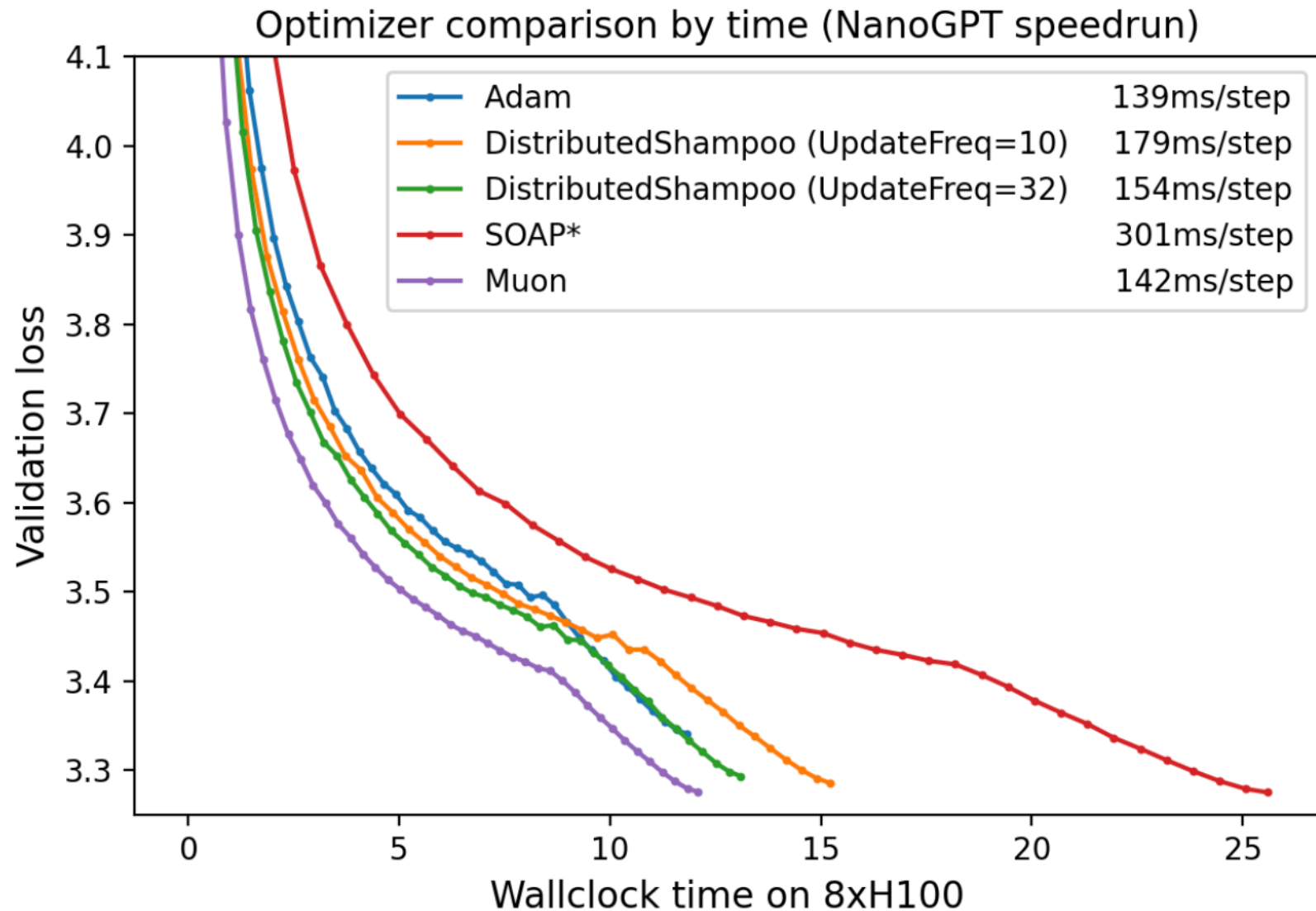
Can "tune" coefficients for specific characteristics.

- Higher-order may converge faster, but each step is more expensive.

Do you have to converge?

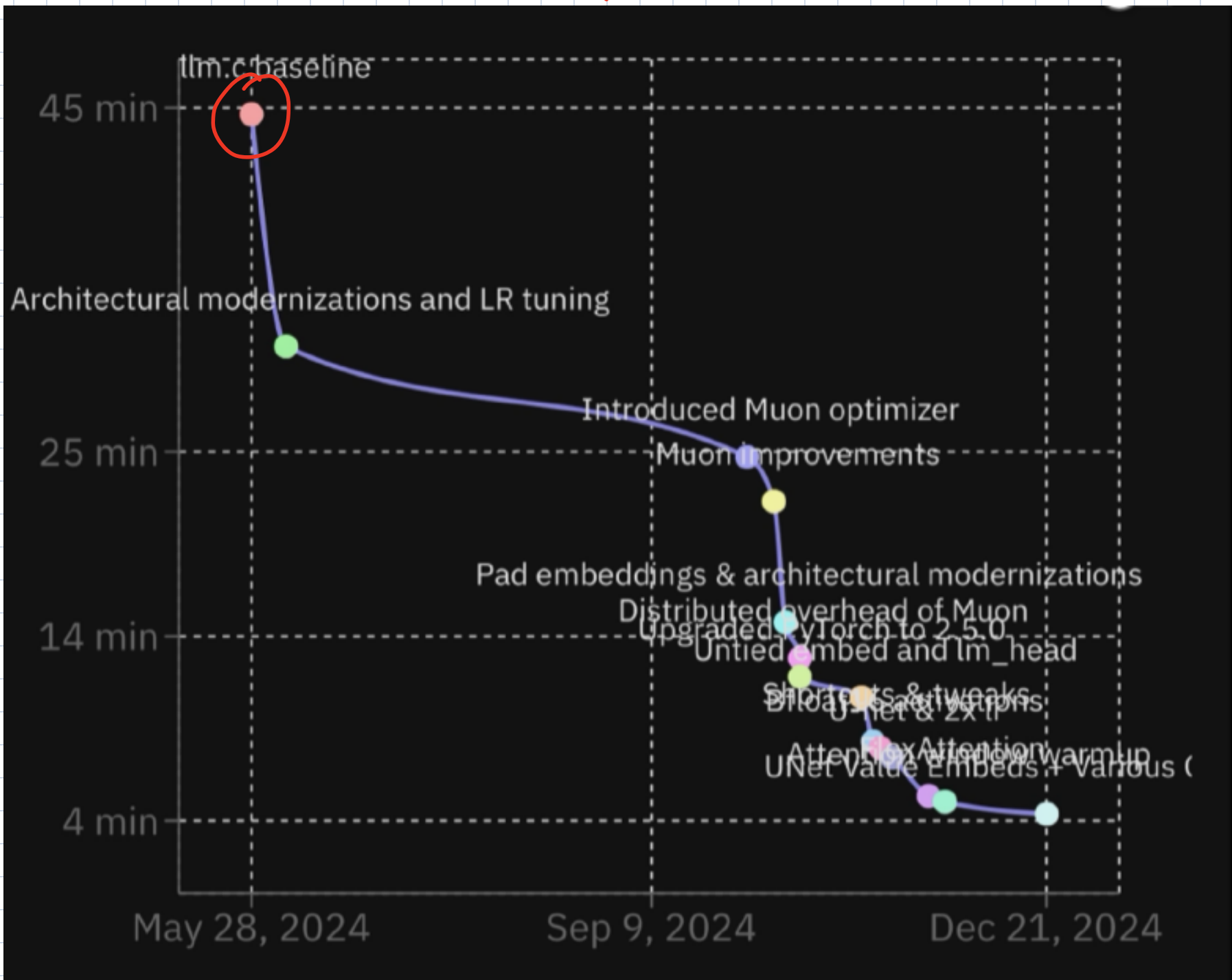
NanoGPT speedrun : $f(x) = 3.444x - 4.7750x^3 + 2.0315x^5$
 $f(1) \neq 1$





*SOAP is under active development. Future versions will significantly improve the wallclock overhead.

NanoGPT speedrun



MUON IS SCALABLE FOR LLM TRAINING

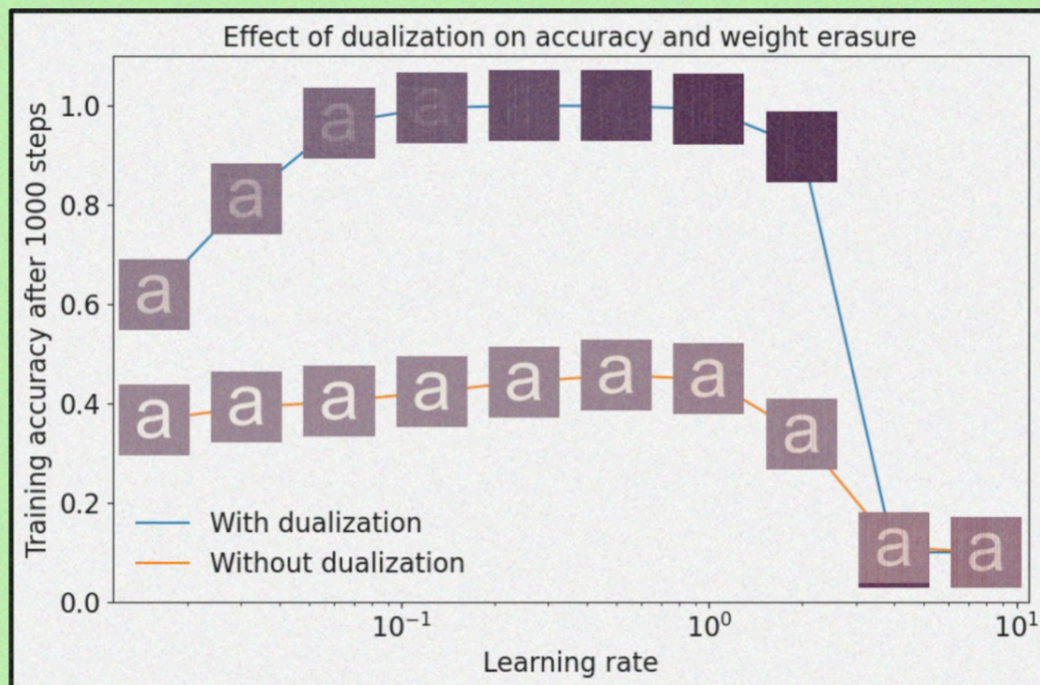
TECHNICAL REPORT

Jingyuan Liu¹ Jianlin Su¹ Xingcheng Yao² Zhejun Jiang¹ Guokun Lai¹ Yulun Du¹
Yidao Qin¹ Weixin Xu¹ Enzhe Lu¹ Junjie Yan¹ Yanru Chen¹ Huabin Zheng¹
Yibo Liu¹ Shaowei Liu¹ Bohong Yin¹ Weiran He¹ Han Zhu¹ Yuzhi Wang¹
Jianzhou Wang¹ Mengnan Dong¹ Zheng Zhang¹ Yongsheng Kang¹ Hao Zhang¹
Xinran Xu¹ Yutao Zhang¹ Yuxin Wu¹ Xinyu Zhou¹ * Zhilin Yang¹

¹ Moonshot AI ² UCLA



22 February 2025



Dualized training converges faster and erases the watermark faster. I wrote up a thorough explanation for this effect in the [Modula docs](#).

So can we use this induced RMS $\rightarrow$ RMS norm in our optimizer recipe?

$$\|\Delta W\|_{\text{RMS}^*} \leq \gamma ?$$

Key observation: If we choose this norm, we can choose only one hyperparameter γ across all layers, but the effect will be layer specific learning rates, due to fan-in and fan-out.

$$\begin{aligned} \|\Delta W\|_{\text{RMS}^*} \leq \gamma &\Rightarrow \sqrt{\frac{d_1}{d_2}} \|\Delta W\|_2 \leq \gamma \\ &\Rightarrow \|\Delta W\|_2 \leq \gamma \sqrt{\frac{d_1}{d_2}} \end{aligned}$$

$\rightarrow$ Essential idea behind mup $\rightarrow$ maximal update parameterization.

Different Approach.

Consider Sign SGD/Adam.

$$W_{t+1} = W_t - \eta \cdot \text{sign}(\nabla_w \mathcal{L}(w)).$$

But I want to rethink stepsize choice

Factor in that different layers have different architectures.

Maybe RMS norm can help?

$$\text{Want: } \|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \delta$$

Consider mini-batch size 1.

$\Rightarrow \nabla_w \mathcal{L}$ gradient is low-rank, specifically rank 1.

$$\Rightarrow \nabla_w \mathcal{L} = \sigma \cdot \vec{u} \vec{v}^T$$

$$\|\nabla_w \mathcal{L}\|_2^2 = \sigma^2$$

$$\|\nabla_w \mathcal{L}\|_F^2 = \text{trace}(\sigma^2 \vec{v} \vec{u}^T \vec{u} \vec{v}^T) = \sigma^2 \quad (\text{sum of singular values squared})$$

Consider $\text{sgn}(\nabla_w \mathcal{L}) = S$. (elementwise)

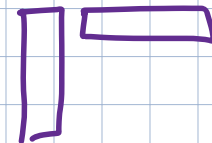
What is rank S ?

$$\text{sgn}(S) = \text{sgn}(\sigma \vec{u} \vec{v}^T)$$

$$= \text{sgn}(\vec{u} \vec{v}^T)$$

$$= \text{sgn}(\vec{u}) \cdot \text{sgn}(\vec{v}^T)$$

$$\text{sgn}(u_i v_j) = \text{sgn}(u_i) \cdot \text{sgn}(v_j)$$



So S is also rank 1.

$$W_{t+1} = W_t - \underbrace{\eta \cdot S}_{\Delta W}$$

Recall, we want ΔW to be small.

$$\| \Delta W \|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma$$

$$\eta \cdot \| S \|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma \quad \Rightarrow \quad \eta \cdot \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \cdot \| S \|_2 \leq \gamma$$

What is $\| S \|_2$?

Recall, we want ΔW to be small.

$$\| \Delta W \|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma$$

$$\eta \cdot \| S \|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma \quad \Rightarrow \quad \eta \cdot \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \cdot \| S \|_2 \leq \gamma$$

What is $\| S \|_2$?

S is rank 1, so spectral norm = Frobenius norm.

$$\| S \|_F^2 = d_{\text{in}} \times d_{\text{out}}$$

Entries are just ± 1 .

$$\Rightarrow \| S \|_2 = \sqrt{d_{\text{in}} \times d_{\text{out}}}$$

$$\Rightarrow \eta \cdot \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \cdot \sqrt{d_{\text{in}} \times d_{\text{out}}} \leq \gamma$$

$$\Rightarrow \eta \leq \frac{1}{d_{\text{in}}} \cdot \gamma$$

Learning rate can scale with layer.

So my Adam / signSGD learning rate should scale as $\frac{1}{d_{in}}$
to maintain $\|\Delta W\|_{RMS \rightarrow RMS} \leq 1$

→ This captures an essence of the μP parameterization
(Maximal update parameterization)
Yang et al. 2022.

Notes:

① As batch size increases, Frobenious norm will only
be an upper bound to spectral norm.

So our bound will be conservative.

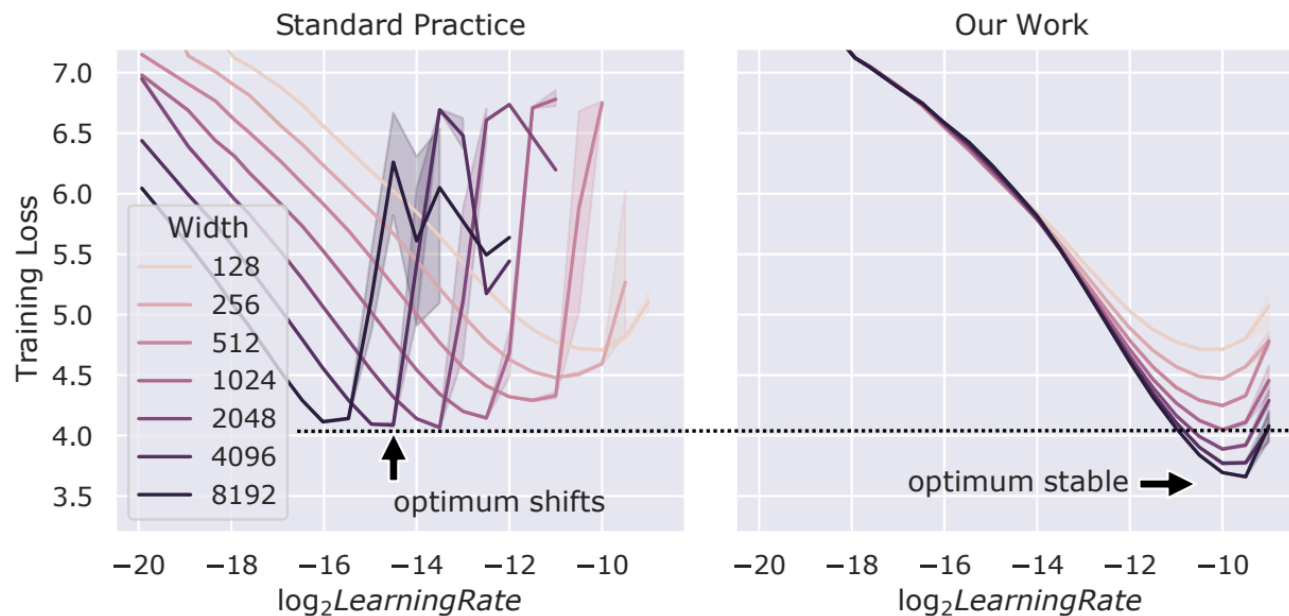


Figure 1: Training loss against learning rate on Transformers of varying d_{model} trained with Adam. Conventionally and in contrast with our technique, different widths do not share the same optimal hyperparameter; wider networks do not always perform better than narrower ones; in fact they underperform the same-width networks in our technique even after tuning learning rate (see dashed line). See [Sections 3](#) and [4](#) for experimental setup.

Yang et al 2022.

What is parameterization? Why does it matter?

* μP :
How to change
hyp. params when NN
width changes.

- About choosing the right "units"
- A key challenge for neural net training is hyperparameter search.
- Particularly bad if the network is large.
 - Experimenting is very expensive.
- Can we optimize hyperparameters on smaller networks and then "transfer" them to larger ones? What is the right scaling?

Example: $x_1, x_2, \dots, x_n$ iid from zero-mean, unit variance distribution.

Then $\frac{1}{\sqrt{n}} (x_1 + x_2 + \dots + x_n) \xrightarrow{n \rightarrow \infty} N(0, 1)$

But $\frac{1}{n} (x_1 + x_2 + \dots + x_n) \xrightarrow{n \rightarrow \infty} 0$

$$\frac{1}{1} (x_1 + x_2 + \dots + x_n) \xrightarrow{n \rightarrow \infty} \infty$$

$\frac{1}{\sqrt{n}}$ is the right order of "scaling factor" so that $G(\sum x_i)$ converges to something non-trivial.

Consider

$$F_n(c) \doteq \mathbb{E}_{x_1, x_2, \dots, x_n} f(C(x_1 + x_2 + \dots + x_n))$$

$c \in \mathbb{R}$, $f: \mathbb{R} \rightarrow \mathbb{R}$ bounded, continuous.

Reparametrize $c = \alpha / \sqrt{n}$.

$$G_n(\alpha) \doteq F_n\left(\frac{\alpha}{\sqrt{n}}\right) \rightarrow \mathbb{E}\left(f(N(0, \alpha^2))\right)$$

as $n \rightarrow \infty$

only function of α .

$\alpha_n^* = \underset{\alpha}{\operatorname{argmin}} G_n(\alpha) \rightarrow$ Then $\alpha_n^* \approx \alpha_N^*$ for large enough n, N

$\rightarrow$ We can copy α_n^* from a smaller n to a larger N .

If instead $c_n^* = \underset{c}{\operatorname{argmin}} F_n(c) \rightarrow$ Cannot transfer parameters in the same way.

Can we do something similar for NN hyperparameters?

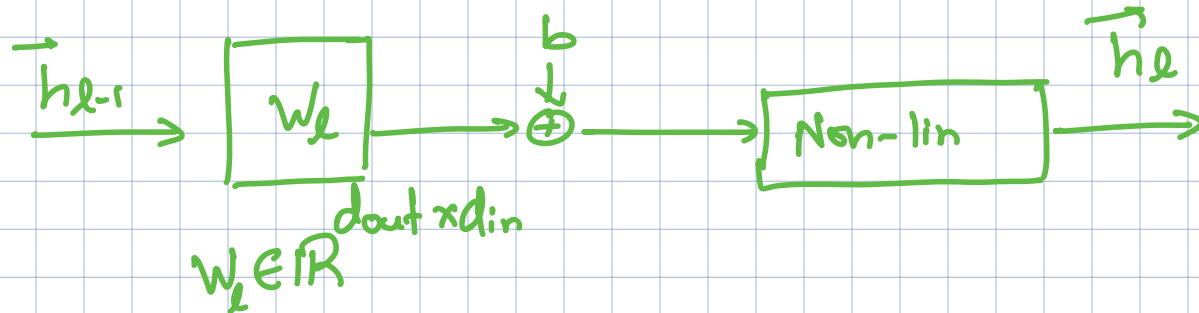
$$\eta \leq \frac{1}{d_{\text{in}}} \cdot ?$$

still a hyperparameter.
the "right scaling".

Conditions for Feature Learning

$\vec{h}_l$: hidden layer l .

$$\vec{h}_{l-1} \in \mathbb{R}^{d_{in}} \quad \vec{h}_l \in \mathbb{R}^{d_{out}}$$



Focus on scaling with network width.

Xavier initialization thinking suggests we want

$$\|\vec{h}_l\|_{\text{RMS}} = \frac{1}{\sqrt{d_{out}}} \|\vec{h}_l\|_2$$

$$\Rightarrow \|\vec{h}_l\|_2 \approx \sqrt{d_{out}}$$

$$\|\vec{h}_l\|_{\text{RMS}} \approx 1.$$

Relaxation of all entries approx 1.

$$\|\vec{h}_l\|_{\text{RMS}} = \Theta(1)$$

Also, want to ensure that our updates to the layers are not negligible as width grows.

$$\|\Delta \vec{h}_l\|_{\text{RMS}} = \Theta(1)$$

Desiderata: (1) $\|\vec{h}_l\|_{\text{RMS}} = \Theta(1)$

(2) $\|\Delta \vec{h}_l\|_{\text{RMS}} = \Theta(1)$

How can we achieve this?

For now, ignore bias + non-linearity.

$$\vec{h}_l = W_l \vec{h}_{l-1}$$

$\Rightarrow$ if $\vec{h}_{l-1}$ aligns with top singular vector of W_l . [Can show this is the case]

$$\|\vec{h}_l\|_2 = \|W_l\|_2 \cdot \|\vec{h}_{l-1}\|_2$$

$$\Theta(\sqrt{d_{out}}) = \|W_l\|_2 \Theta(\sqrt{d_{in}})$$

$$\Rightarrow \|W_l\|_2 = \Theta\left(\sqrt{\frac{d_{out}}{d_{in}}}\right) \quad \text{i.e. } \|W_l\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$$

i.e. fixed RMS norm as we started with!

Thus, if $\|W_l\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$ and $\|\vec{h}_{l-1}\|_{\text{RMS}} = \Theta(1)$, then

$$\|\vec{h}_l\|_{\text{RMS}} = \Theta(1)$$

What about $\|\Delta h_l\|_{\text{RMS}}$?

$\rightarrow$ Need one more condition.

$$\| \Delta W_e \|_{\text{RMS}} = \Theta(1) \Rightarrow \| \Delta W_e \|_2 = \Theta \left(\sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} \right)$$

$$\underline{\vec{h}_e = W_e \vec{h}_{e-1}}$$

$$\Rightarrow \Delta \vec{h}_e = \Delta W_e \cdot \vec{h}_{e-1} + W_e \cdot \Delta \vec{h}_{e-1}$$

$$\begin{aligned} \Rightarrow \| \Delta \vec{h}_e \|_2 &\leq \| \Delta W_e \|_2 \cdot \| \vec{h}_{e-1} \|_2 + \| W_e \|_2 \| \Delta \vec{h}_{e-1} \|_2 \\ &= \boxed{?} \Theta(\sqrt{d_{\text{in}}}) + \Theta \left(\sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} \right) \cdot \Theta(\sqrt{d_{\text{in}}}) \end{aligned}$$

$$\Rightarrow \| \Delta h_e \|_{\text{RMS}} = \Theta(1)$$

Conditions: (1) $\| W \|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$

(2) $\| \Delta W \|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$

$$\vec{\Delta h}_\ell = \Delta W_\ell \cdot \vec{h}_{\ell-1} + W_\ell \cdot \vec{\Delta h}_{\ell-1}$$

$$\begin{aligned} \|\vec{\Delta h}_\ell\|_2 &= \|\Delta W \vec{h}_{\ell-1} + W_\ell \vec{\Delta h}_{\ell-1}\|_2 \\ &\leq \|\Delta W\|_2 \|\vec{h}_{\ell-1}\|_2 + \|W_\ell\|_2 \|\vec{\Delta h}_{\ell-1}\|_2 \end{aligned}$$

→ apply triangle inequality
+ sub-multiplicativity property